

A PROJECT REPORT ON
Talk2Invoice: AI-Powered Voice Assistant for Food Ordering
Using Whisper and DistilBERT

Submitted in partial fulfilment for the award of the degree of
BACHELOR OF TECHNOLOGY

In
Artificial Intelligence and Machine Learning

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ABSTRACT

Artificial intelligence is being adopted more in fast-food service environments to make ordering quicker and easier, especially in busy places like metro stations, airports, and self-service kiosks. However, many current solutions don't meet expectations due to limited voice recognition, lack of real-time inventory updates, and inadequate backend systems. Talk2Invoice is a voice assistant powered by AI that aims to solve these issues by allowing contactless food ordering through natural speech and automating backend tasks. Multilingual voice input is turned into text using OpenAI's Whisper model, and customer intent along with food item details is identified using a fine-tuned DistilBERT model. QR code-based menu access makes interactions simpler, while a blockchain-enabled backend supports real-time stock management, queue optimization, and secure digital transactions. Once the order is confirmed successfully, digital invoices are automatically generated and sent to users, and analytical reports provide insights into sales trends and service performance. Experimental results show improved order accuracy in noisy settings, and better transparency in transaction handling. By combining speech processing, natural language understanding, and secure backend technologies, Talk2Invoice presents a practical, scalable, and user-friendly method for AI-driven food ordering in modern service environments.

Keywords: *Voice Assistant, Speech-to-Text, Whisper (Faster Whisper), DistilBERT, Natural Language Processing, Real-Time Inventory Management, Digital Invoicing, Blockchain*

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CHAPTER-1

INTRODUCTION

1. INTRODUCTION

1.1 Introduction

Artificial intelligence is reshaping service industries by enabling smart automation and natural human–computer interaction. In fast-food and quick-service restaurant environments, particularly in high-traffic locations such as airports, metro stations, shopping malls, food courts, and self-service kiosks, efficient order placement and transaction processing are essential for maintaining service quality and customer satisfaction. Traditional food ordering systems, including manual counters and touchscreen kiosks, often lead to long waiting lines, incorrect orders, and reduced operational efficiency during peak hours. These challenges create difficulties for both customers and restaurant staff in managing high demand effectively. As a result, there is a growing need for AI-driven solutions that can streamline ordering processes, reduce human errors, and enhance the overall dining experience. Voice-based interfaces have emerged as a promising alternative by providing a convenient, contactless, and user-friendly method for customers to place food orders using natural speech. However, existing voice assistants often face limitations in real-world fast-food environments due to background noise, multilingual conversations, and limited integration with backend systems such as billing, inventory, and order management platforms. These limitations frequently result in incorrect intent recognition, delayed responses, and inconsistent order fulfilment. To address these challenges, this study introduces Talk2Invoice, an AI-powered voice assistant designed to automate food ordering and invoicing through natural speech interaction. The system aims to improve order accuracy, reduce service time, and ensure seamless integration with backend restaurant systems. By combining natural language understanding, intelligent intent recognition, and automated billing mechanisms, Talk2Invoice provides a scalable and reliable solution for modern fast-food service environments.

The Need for Voice-Based Food Ordering Automation

Fast-food environments need quick and precise order processing to handle high customer volumes effectively. Manual and touchscreen systems often cause delays and errors, especially during busy times. Additionally, the growing demand for contactless interactions has made it necessary to have ordering methods that minimize physical

contact and are easy to use. Voice-based food ordering offers a natural solution, but its effectiveness relies on strong speech recognition and seamless backend integration. There is a clear need for intelligent voice-driven platforms that combine ordering, inventory checks, and billing into a single automated workflow.

1.2 Motivation

In high-traffic service environments like fast-food outlets and airport counters, customers demand quick, contactless, and error-free order processing. Traditional manual or touchscreen systems often cause delays and errors, especially during peak hours. Existing voice assistants also fall short due to poor performance in noisy, multilingual settings and lack of integration with backend systems, leading to latency and billing inaccuracies. To overcome these issues, Talk2Invoice introduces a unified framework combining Whisper for multilingual speech-to-text, DistilBERT for contextual intent recognition, and a blockchain-based backend for secure, automated invoicing. This integration ensures real-time, low-latency, and transparent order processing, addressing both operational efficiency and data integrity in dynamic service environments.

1.3 Scope

The Talk2Invoice system is designed to revolutionize the food service automation process by integrating speech recognition, natural language understanding, and secure digital invoicing into a unified, AI-driven platform. The scope of this work includes the development of a multilingual, real-time, and adaptive voice assistant capable of automating order placement, payment, and analytics in fast-paced environments such as kiosks and restaurants. The project primarily focuses on the following objectives:

Multilingual Speech-to-Text Conversion:

Utilizing OpenAI's Whisper model, the system accurately transcribes spoken input across multiple languages and noisy conditions, ensuring smooth interaction in diverse customer environments. It supports real-time transcription with high precision, enabling faster and more natural communication. The model is capable of handling different accents and speech variations, improving reliability in practical scenarios. This functionality enhances accessibility and ensures a seamless user experience.

Dynamic Inventory Management:

The system continuously tracks stock availability, automatically filtering unavailable items and optimizing kitchen load distribution through real-time analytics.

Selling Soon-to-Expire Food:

Through AI-based inventory monitoring, the system identifies ingredients nearing expiration and promotes them via discounted “Smart Menu” items, minimizing food waste and improving sustainability.

CHAPTER-2

LITERATURE SURVEY

2. LITERATURE SURVEY

Recent developments in Artificial Intelligence, Natural Language Processing, and Speech Recognition have played an important role in creating automated voice-based ordering systems for restaurants and self-service environments. Srinivasan et al. [1] proposed a voice assistant for fast-food restaurants using the DistilBERT model to improve the accuracy of speech recognition and intent detection. Their work mainly focused on reducing response time while keeping high performance in language processing. Kim et al. [2] introduced a voice assistance system that supports real-time speech processing and smooth communication between the user input and backend services. Their study showed that reliable speech recognition is very important for building interactive applications. In a similar way, Zhang et al. [3] developed an intelligent voice-based food ordering system that combines speech recognition with backend order management, allowing customers to place orders without manual interaction. This helps in reducing human effort and increases service speed in restaurants.

Speech-based web applications have also been studied for improving restaurant automation. Shirode et al. [4] designed a speech-oriented virtual restaurant clerk using the Web Speech API along with Natural Language Processing techniques. Their system allows users to place food orders using voice commands through a web browser, making it useful for kiosks and self-service machines. Kumar et al. [5] provided a detailed review of voice assistants used in daily life and explained their applications in customer service, smart devices, and automated systems. The study pointed out that voice assistants make systems easier to use, but they still have problems in understanding different accents and working in noisy environments. Ashfaq et al. [6] studied the use of chatbots in online food delivery services and compared social-oriented and task-oriented interaction styles. The results showed that task-oriented chatbots work better for food ordering because they give faster and more accurate responses.

Many researchers have also developed voice-enabled web and smart ordering applications to improve user experience and automation. Patil et al. [7] created a voice-assisted food ordering web application called Vocalicious, which allows users

to order food using speech commands through a simple interface. Their system focused on making the ordering process easy and user-friendly. Rathod et al. [8] also developed a speech-based virtual restaurant clerk using speech recognition and Natural Language Processing to support hands-free ordering in restaurants. Huang et al. [9] proposed a hybrid method that combines rule-based techniques with modern language models for on-device food ordering systems. This method improves system reliability and reduces the need for cloud processing, making the system faster. Alharbi et al. [10] developed a smart restaurant ordering system using deep learning and Internet of Things (IoT) technology. Their system connects voice input, smart devices, and backend services to automate the entire ordering process, showing how intelligent automation can improve the food service industry.

Even though many systems have been developed, some limitations still exist. Most voice-based ordering systems mainly focus on speech recognition and chatbot interaction, but they do not provide proper real-time inventory updates, multilingual support, or strong backend integration with billing and order management systems. In addition, some systems depend too much on cloud processing, which can increase response time. Because of these limitations, there is a need to develop a better AI-powered voice ordering system that supports natural conversation, real-time data processing, multiple languages, and efficient backend integration. Such a system can provide a faster, contactless, and more user-friendly ordering experience in modern fast-food restaurants and self-service environments.

CHAPTER-3

SYSTEM STUDY AND ANALYSIS

3. SYSTEM STUDY AND ANALYSIS

3.1 Problem Statement

In fast-food places like restaurants, airports, and kiosks, customers often face delays and confusion while placing orders using manual or touchscreen systems. These systems are slow, cause long queues, and lead to mistakes during busy hours. Existing voice assistants also have problems — they cannot work well in noisy environments, do not understand different languages properly, and are not connected to real-time inventory or billing systems. This leads to wrong orders and poor customer experience. To solve these issues, there is a need for a smart voice-based system that can take food orders using natural speech, understand customer intent, check available items, and automatically generate bills. **Talk2Invoice** aims to provide this solution by using artificial intelligence to make food ordering faster, easier, and more accurate.

3.2 Existing System

In the current fast-food ordering systems, most restaurants use manual entry or touchscreen methods to take customer orders. These systems are often slow and can lead to mistakes, especially during rush hours. Some voice assistants are used, but they are not very effective. They work only with limited voice commands, do not understand different languages properly, and cannot handle background noise. Also, they are not fully connected to other systems like inventory and billing. Because of this, staff still need to check stock and prepare bills manually, which causes delays and errors. Overall, the existing systems are not smart enough to provide fast, accurate, and fully automated food ordering.

3.3 Limitations of the Existing System

1. Voice assistants do not work well in noisy places.
2. They cannot understand different languages or natural speech properly.
3. There is no emotion or mood detection to make conversations more natural.
4. The systems are slow and not suitable for busy, high-traffic areas.

3.4 Proposed System

The Talk2Invoice system is an AI-powered voice assistant that streamlines food ordering by integrating speech recognition, NLP, and blockchain technology for fast, secure, and reliable operations. Customers give voice commands through a kiosk or microphone, which are processed by Fast Whisper and OpenAI’s Whisper models to convert multilingual speech into accurate text, even in noisy environments. The transcribed text is then analyzed by a fine-tuned DistilBERT model to understand customer intent—such as placing, modifying, or checking orders—and to extract specific items and quantities. The system connects directly to the restaurant’s inventory database to verify item availability, suggest alternatives, manage kitchen load, and promote near-expiry items as special offers. Confirmed orders are securely recorded on a blockchain-based backend, ensuring transparency and data integrity through smart contracts that automate billing, payments, and invoice generation. Finally, the system delivers a digital invoice via QR code, SMS, or email and provides analytics reports on sales and performance. Overall, Talk2Invoice offers a voice-driven, low-latency, and secure solution for modern food service automation.

3.5 Functional Requirements

The functional requirements define the core operations and behaviors expected from the *Talk2Invoice* system. Each requirement supports the end-to-end automation process, from voice input to invoice generation.

- **Voice Order Collection:**

The system must capture real-time user voice input and record spoken orders and preferences accurately through the built-in microphone or input device.

- **Speech-to-Text Processing:**

The *Whisper* model shall be employed to convert multilingual speech into text with high accuracy, ensuring effective operation even in noisy environments such as food courts or metro stations.

- **Intent Recognition:**

The system must utilize the *DistilBERT* model to identify user intent and extract specific entities such as food items, quantity, and modifiers from the transcribed

- **Inventory and Load Check:**

The system shall verify the real-time availability of food items and balance kitchen workloads dynamically to reduce service delays and avoid overloading specific preparation units.

- **Blockchain Integration:**

The system shall verify the real-time availability of food items and balance kitchen workloads dynamically to reduce.

- **Order Confirmation and Payment:**

Upon successful intent recognition and inventory verification, the system shall confirm the order with the user and facilitate secure QR-based payment processing.

- **Invoice Generation:**

The system should automatically generate digital invoices linked to blockchain transaction records and deliver them to users via email or QR code.

- **Analytics and Continuous Learning:**

The backend should produce daily analytical reports summarizing order statistics, performance metrics, and stock levels. It must also use user feedback to enhance future recognition accuracy and system responsiveness.

3.6 Non-Functional Requirements

The non-functional requirements specify the performance, reliability, and quality standards that the

Talk2Invoice system must achieve to ensure optimal operation in real-world environments.

- **Performance:**

The system should handle multiple concurrent voice orders with minimal latency, providing near real-time transcription and intent recognition.

- **Security:**

All user and transaction data must be encrypted, while the blockchain ensures tamper-proof order verification and secure digital invoicing

- **Reliability:**

The system shall maintain stable operation under high user load without crashes or data loss, ensuring consistent service availability.

- **Usability:**

The interface must be intuitive, allowing customers to interact naturally through voice commands without requiring prior technical knowledge.

3.7 Software Requirements

The Talk2Invoice platform requires a combination of AI, NLP, web, and blockchain technologies to function efficiently. The following software tools and frameworks are essential for system development and deployment:

- **Programming Language:**

Python — for core logic, machine learning model integration, and backend development. Python provides high flexibility and strong library support, making it suitable for integrating speech recognition, natural language processing, and backend services in a single system.

- **Libraries and Frameworks:**

OpenAI Whisper, HuggingFace Transformers (for DistilBERT), Flask or FastAPI (for RESTful backend), scikit-learn (for preprocessing), Web3.py (for blockchain interaction), and Matplotlib (for analytics visualization).

- **Development Environment**

Jupyter Notebook, Google Colab, Visual Studio Code, or PyCharm for training, testing, and deploying models.

- **Database:**

MongoDB — for storing user profiles, order details, inventory records, and transaction histories. MongoDB is used as the database to store user details, order information, inventory data, and transaction records in a structured and secure way. Its flexible and scalable design allows the system to handle real-time updates efficiently and manage restaurant data without performance issues.

3.8 Hardware Requirements

Efficient execution of speech recognition, NLP models, and blockchain operations requires sufficient computational and network infrastructure. The following hardware specifications are recommended:

- **Processor:**

Multi-core CPU or GPU (NVIDIA or AMD) to accelerate Whisper transcription and DistilBERT inference tasks.

- **Memory (RAM):**

Minimum of 8 GB; 16 GB or higher is recommended for real-time parallel processing and model execution.

- **Networking:**

Stable broadband or wireless internet connection for API communication and cloud-based inference .Local LAN connectivity for kiosk deployment in high-traffic locations such as metro stations or airport terminals.

CHAPTER-4

SYSTEM DESIGN

4. SYSTEM DESIGN

4.1 System Architecture

System architecture is a conceptual model that defines the structure, behavior, and interactions of a system's components, acting as a blueprint for development. It outlines how hardware, software, data, and users work together to meet technical and business goals, ensuring scalability, performance, and security.

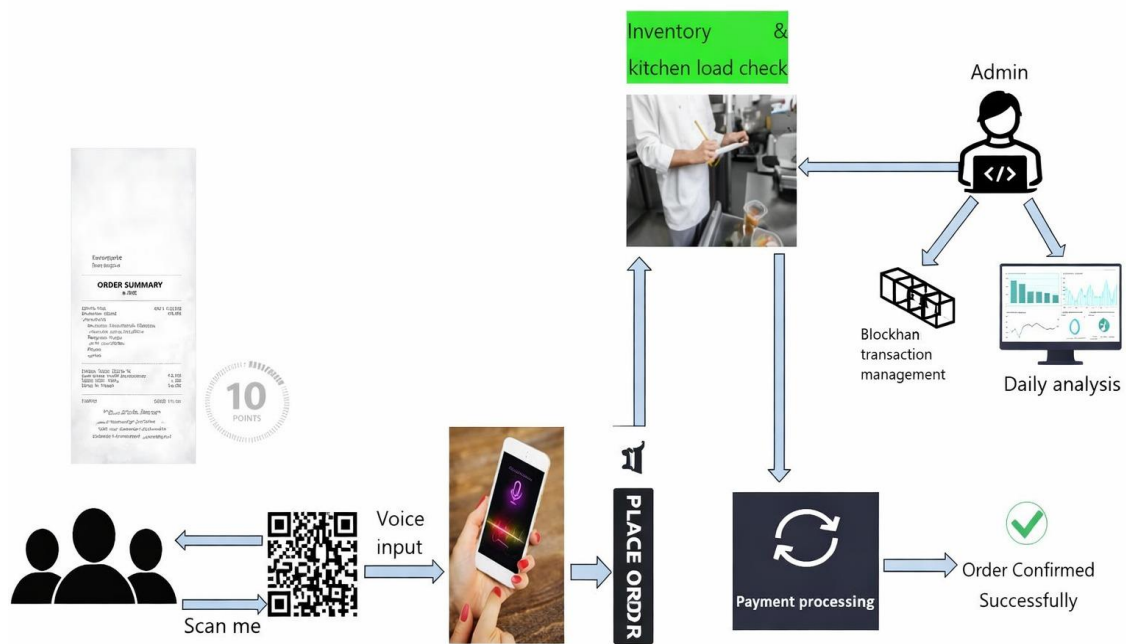


Fig. 4.1: System Architecture

The Talk2Invoice system starts with the user, who gives a voice command to order food (for example, “I want a pizza and coke”). The system listens to the user’s voice and converts it into text using the Whisper and Fast Whisper models. Once converted, the text is analyzed by the DistilBERT model, which understands what the user wants to order and the quantity. The system then checks the menu to see if the requested items are available. It sends this information to the cart and also updates the inventory accordingly. The user’s selected items are added to the cart, where they can review or modify their order before confirming it. After the order is finalized, the system calculates the total amount and generates an invoice, which is sent to the user through a QR code,

The admin manages the entire system — updating menus, checking inventory, and monitoring reports. The inventory management module keeps track of available food items and automatically updates stock after each order. Finally, the dashboard displays daily sales, total orders, and performance reports, helping the admin monitor and manage the system efficiently

4.2 System Design

System design is the process of defining the components, modules, interfaces, and data flow of a system to meet specific requirements. It focuses on how different parts of the system work together to achieve the desired functionality

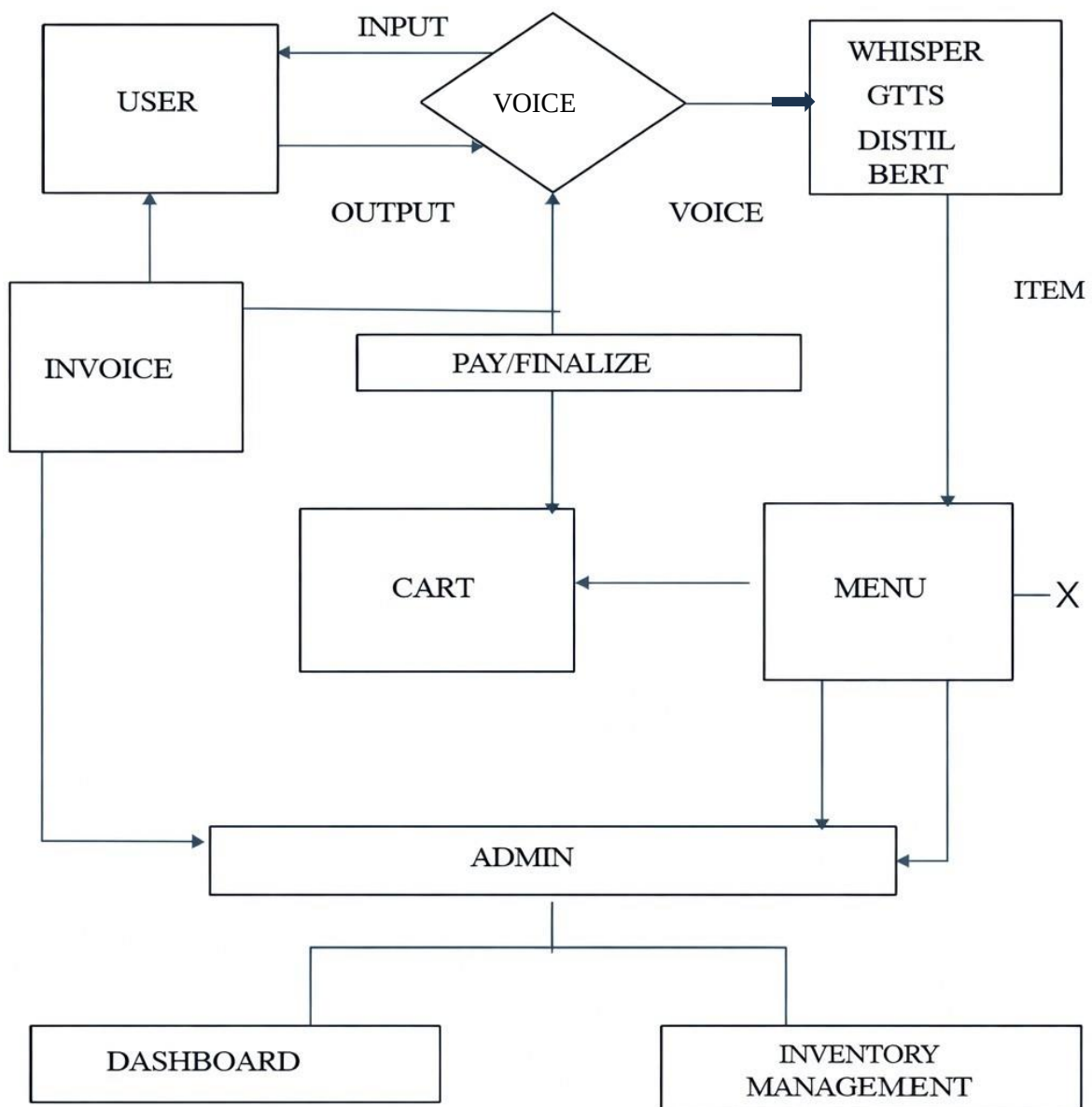


Fig. 4.2: System Design

The diagram shows how the Talk2Invoice system works from start to finish

- **User and Voice Input:**

The process begins with the user, who uses a smartphone or device to give a voice command (for example, “Order one burger and a coke”).

- **Voice to Order Placement:**

The system listens to the voice input and processes it using Whisper and Fast Whisper to convert the speech into text. Then, DistilBERT analyzes the text to understand what items the user wants to order. Once recognized, the system allows the user to place the order.

- **Inventory & Kitchen Load Check:**

After the order is placed, the system checks the inventory to confirm if the requested items are available and also monitors kitchen workload. If the items are available, the order moves forward; otherwise, alternative suggestions can be provided.

- **Payment Processing:**

Once the inventory confirms availability, the system proceeds to payment processing. Users can pay digitally, and the transaction details are handled securely.

- **Blockchain Transaction Management:**

The order and payment details are recorded securely in a blockchain-based backend. This ensures that all transactions are transparent, verified, and cannot be tampered with.

- **Admin & Daily Analysis:**

The admin monitors the system through a dashboard. They can view orders, manage inventory, and analyze daily sales and performance reports generated by the system.

- **Order Confirmation:**

After payment and verification, the system confirms the order. A digital invoice is generated and linked to a QR code for the user to scan and check details. The user is notified that the order is confirmed successfully.

- **Invoice Generation:**

The customer can view or download the digital bill that contains order details, payment confirmation, and estimated preparation time.

CHAPTER-5

METHODOLOGY

5. METHODOLOGY

The methodology of the Talk2Invoice system focuses on a voice-based approach for food ordering and invoice generation. It consists of three main components: voice interaction and intent processing, order validation, and backend management. This structured approach ensures smooth workflow from customer interaction to administrative control. Overall, it improves reliability and efficiency in restaurant operations.

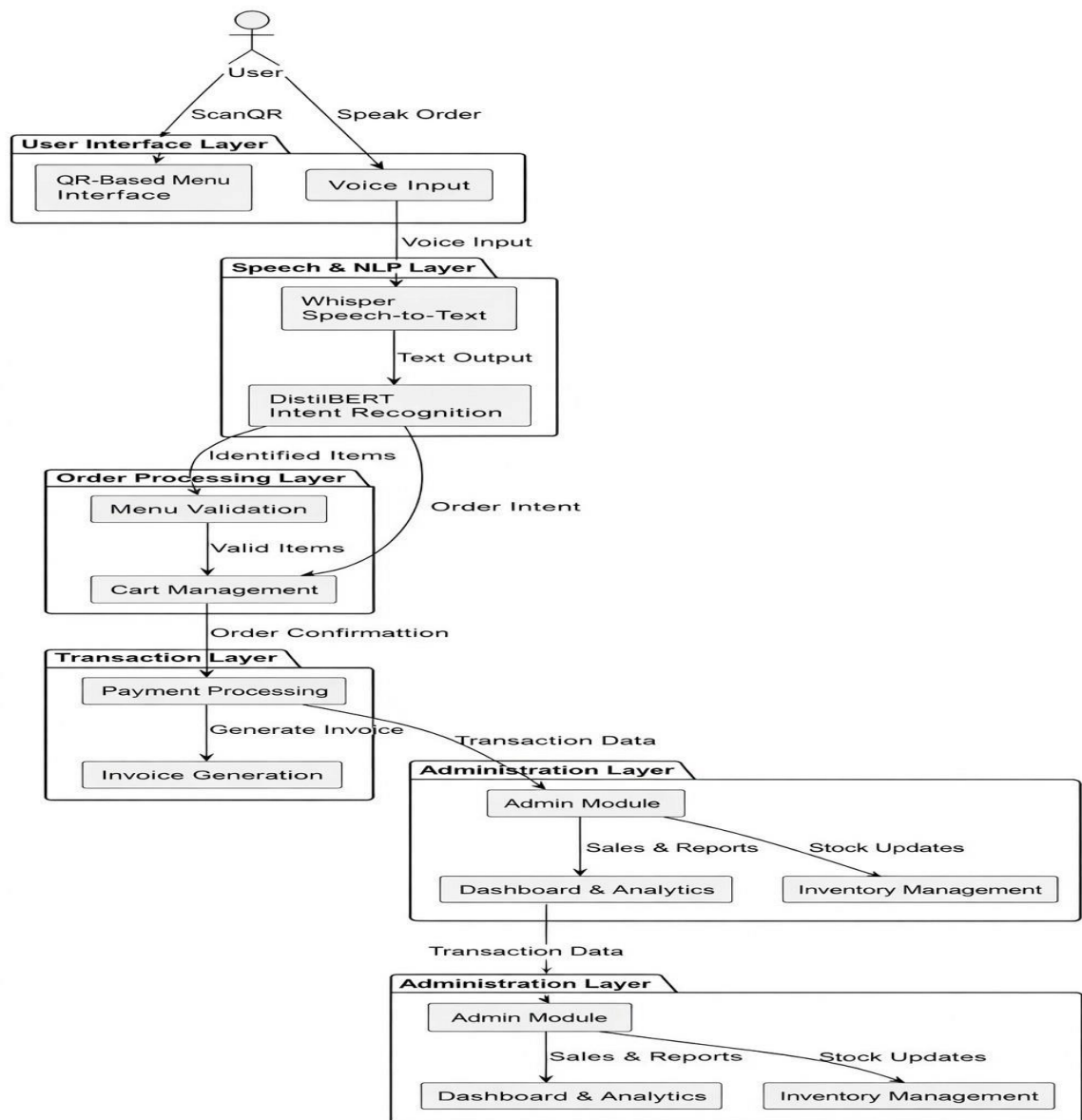


Fig. 5.1: Methodology Flow Chart

5.1 Voice interaction and intent processing process

This begins with a customer scanning a QR code in the restaurant which directs to a digital menu interface. The interface is browser-based and is easy and it allows the customers to use the system without the need to install an app. There is also an interactive element of voice in the menu. The customer orders a product after talking to a device that has a microphone. Background noise or other speech patterns may be picked up by the recorded voice which is common in restaurants. This sound is sent to a Speech-to-Text (STT) engine which translates the request to text using the Whisper or Faster-Whisper engine. Through the Intent Recognition module, the text is then run through a fine-tuned DistilBERT model. This model examines the context in order to determine what the user wants and extract appropriate information such as the name and quantities of menu items. The information that was interpreted is added to the cart temporarily and a voice confirmation message is issued by the system to enable the customer check the detected order before proceeding.

5.2 Processing and validation of orders

Once the user affirms with the order observed, the system initiates the order processing stage. The backend performs feasibility checks prior to the acceptance of the order to make sure that nothing will go wrong. The system determines the inventory first to identify whether the required items in stock. Meanwhile, it evaluates the workload of the kitchen so that the delays are not long, in particular, in peak times. Only the orders which comply with both the inventory and kitchen workload requirements are further processed. After the validation, the order is completed and assigned with a unique identifier, which aids in the tracking and reference during the life of the order. This validation process checks both inventory availability and kitchen workload before confirming the order to avoid errors and delays. By verifying these conditions in advance, the system ensures accurate order processing even during peak hours.

5.3 Administrative and Backend Management

Once the order is confirmed, the backend system safely stores the details of the order in database to track and keep the records of the order. Also, order and transaction metadata are stored in a blockchain-based ledger which ensures secure storage, immutability and transparency of records. This configuration facilitates in auditing, traceability and dependable recording of transactions. Once the order has been processed, a digital invoice is generated, which contains the details of the items and their quantities and prices and references of transactions. This is e-mailed to the customer. There is an administrative monitoring interface where the restaurant personnel can monitor the manner in which the systems operate. With this interface, administrators are able to monitor the orders placed and orders fulfilled, control stored inventory, monitor kitchen workload and finally check on the transaction record. This centralized management facilitates proper control and proper management of the restaurant.

5.4 Model Training and Evaluation

Talk2Invoice system has been trained and tested through a comprehensive method to make sure that the intent recognition is correct and the performance is reliable in real-time.

Model Training The intent recognition component was realized with the help of a fine-tuned DistilBERT model that was trained on a selection of food item descriptions and utterances related to orders. The DistilBERT tokenizer was used to tokenize the text and Adam W was used to train with a batch size of 32 and a range of learning rates between 2×10^{-5} to 5×10^{-5} . Monitoring validation accuracy and loss during training was done and to prevent overfitting early stopping was used. Inference and Post Processing In inference the Whisper speech recognition model was used to first translate spoken user input into text. The output of that transcription was fed into the trained DistilBERT classifier to determine the intent of the user and the types of foods they wanted, which were converted to backend operations

Model Evaluation Accuracy and Macro-averaged F1-Score were used in evaluating model performance over a held-out test set. The intent recognition module had a maximum accuracy of 96.30%.

In conclusion, the proposed methodology provides a clear and reliable workflow for the Talk2Invoice system, from voice interaction to order processing and backend management. The use of speech recognition, intent detection, and secure data handling improves accuracy and efficiency in restaurant operations. This approach makes the system fast, user-friendly, and suitable for modern contactless food ordering environments.

CHAPTER-6

TECHNOLOGIES

6. TECHNOLOGIES

Technologies form the backbone of the Talk2Invoice system by enabling efficient design, development, and deployment of an intelligent voice-based ordering platform. The system integrates multiple advanced technologies, including programming languages, data processing frameworks, natural language processing models, deep learning architectures, speech recognition systems, and document generation tools. These technologies work collaboratively in a pipeline to convert user voice input into meaningful actions and generate automated invoices in real time.

The integration of AI-driven models with robust software frameworks ensures high accuracy, scalability, and reliability. It also enhances user experience by enabling seamless, fast, and interactive communication between the user and the system. Overall, the technology stack is designed to support real-time processing, intelligent decision-making, and efficient system performance.

6.1 Programming Language and Environment

The Talk2Invoice system is developed using Python, a powerful and flexible programming language widely used in artificial intelligence, machine learning, and data-driven applications. Python's simple and readable syntax improves code maintainability and reduces development complexity, making it highly suitable for rapid prototyping as well as large-scale system development.

Python supports multiple programming paradigms, including object-oriented, procedural, and functional programming, allowing developers to design modular and reusable components. Being an interpreted language, Python facilitates faster debugging, testing, and iteration during development. Its cross-platform nature ensures that the system can run seamlessly on operating systems such as Windows, Linux, and macOS.

One of the key strengths of Python is its extensive ecosystem of libraries and frameworks, such as TensorFlow, NumPy, Pandas, PyTorch, and Transformers, which are essential for implementing AI and data processing functionalities. Python also supports dynamic memory allocation and easy integration with lower-level languages like C and C++, enabling performance optimization when needed. Furthermore, Python supports GUI and web-based development.

6.2 Data Processing and Analysis

Efficient data handling is crucial for managing menu details, customer orders, pricing information, and invoice records within the Talk2Invoice system. The system uses Pandas and NumPy libraries to perform data processing and analysis tasks efficiently and accurately. Pandas provides high-level data structures such as DataFrames, which allow structured data to be stored, accessed, and manipulated in an organized manner. It supports operations like filtering, sorting, grouping, and aggregation, which are essential for handling dynamic order data and updating system records in real time. This ensures that the system maintains consistency and accuracy in all transactions.

NumPy is used for numerical computations and handling multi-dimensional arrays. It provides optimized mathematical operations that significantly improve performance, especially when dealing with large datasets or performing backend calculations. This makes it an essential component for efficient system processing.

JSON is used as a lightweight and flexible data exchange format that enables smooth communication between different system modules. It ensures that structured data such as user orders, responses, and invoice details can be easily transmitted, stored, and processed across various components of the system.

6.3 Natural Language Processing and Deep Learning

Natural Language Processing (NLP) and deep learning techniques form the core intelligence of the Talk2Invoice system. These technologies enable the system to understand human language, interpret user intent, and convert it into actionable commands. This capability allows users to interact with the system naturally using voice or text input.

6.3.1 DistilBERT Model

DistilBERT is a compressed and efficient version of the BERT model that retains most of its powerful language understanding capabilities while significantly reducing computational requirements. It is based on transformer architecture, which uses attention mechanisms to identify relationships between words in a sentence and understand contextual meaning.

The model processes input text bidirectionally, meaning it considers both previous and next words in a sentence to accurately interpret meaning. In the Talk2Invoice system, DistilBERT is used to classify user intent, such as placing an order, modifying items, or

canceling requests. This ensures that the system responds appropriately to different user commands. One of the major advantages of DistilBERT is its balance between performance and efficiency. It delivers high accuracy while maintaining low latency, making it suitable for real-time applications. Its reduced size also allows it to run efficiently on systems with limited computational resources, making it practical for deployment.

6.3.2 Transformers Library (Hugging Face)

The Hugging Face Transformers library simplifies the implementation of NLP models by providing access to a wide range of pre-trained models and tools. It supports tokenization, model loading, fine-tuning, and inference, making it easier to integrate advanced NLP capabilities into the system.

This library reduces development effort and time by eliminating the need to build models from scratch. It also allows customization and fine-tuning of models based on specific application requirements, such as understanding food ordering commands and conversational queries.

6.3.3 Deep Learning Framework (PyTorch)

PyTorch serves as the core deep learning framework used for building, training, and deploying machine learning models in the system. It provides a flexible and dynamic environment that supports experimentation and rapid development. Its dynamic computation graph allows developers to modify and debug models during runtime, which is particularly useful for research and iterative improvements. PyTorch also supports GPU acceleration, which significantly enhances the speed of training and inference, ensuring efficient model performance in real-time applications. PyTorch also provides strong support for integrating with NLP and speech processing libraries, making it suitable for AI-based applications like Talk2Invoice. Its easy-to-use API helps in implementing complex models such as DistilBERT with less code and better readability. This makes PyTorch a reliable choice for developing intelligent, real-time systems where speed, accuracy, and flexibility are important.

6.4 Speech Processing Technologies

Speech processing technologies enable natural and intuitive interaction between users and the Talk2Invoice system. These technologies convert spoken input into text and generate voice responses, enabling a hands-free and user-friendly experience.

6.4.1 Whisper (Speech-to-Text)

Whisper is a deep learning-based speech recognition model developed by OpenAI. It is designed to convert spoken language into text with high accuracy, even in challenging environments with background noise.

The model is trained on large-scale multilingual datasets, enabling it to understand different accents, dialects, and languages. In the Talk2Invoice system, Whisper captures user voice input and converts it into text, which is then processed by the NLP module to identify user intent. This ensures accurate and reliable voice recognition in real-world scenarios.

6.4.2 Fast Whisper

Fast Whisper is an optimized version of the Whisper model that enhances processing speed and reduces latency. It enables faster transcription of speech input, making it ideal for real-time applications where quick responses are essential. By improving efficiency and reducing computational load, Fast Whisper ensures that the system remains responsive and performs well even on devices with limited hardware capabilities.

6.4.3 Text-to-Speech (TTS)

Text-to-Speech (TTS) technology converts system-generated text responses into audible speech, allowing the system to communicate with users effectively. It enhances the user experience by providing real-time feedback, confirmations, and updates through voice output. This technology plays a crucial role in creating an interactive system that feels natural and easy to use, especially in environments where manual interaction is limited.

6.4.4 Google Text-to-Speech (gTTS)

Google Text-to-Speech (gTTS) is used to generate high-quality, natural-sounding voice output from text. It supports multiple languages and produces clear and human-like speech, making it suitable for diverse user groups. In the Talk2Invoice system, gTTS ensures effective communication by delivering system responses in a clear and understandable manner, thereby improving user satisfaction and interaction quality. It also enables real-time audio feedback, helping users receive instant confirmations and system updates.

6.5 Invoice Generation and System Integration

After processing user orders, the system automatically generates digital invoices that include item details, quantities, prices, and total cost. This is achieved using ReportLab, a Python library designed for creating PDF documents programmatically.

ReportLab enables the generation of well-structured, professional invoices that can be stored digitally or shared with customers via email or QR codes. This automation reduces manual effort, minimizes errors, and improves billing efficiency.

To enhance conversational capabilities, the system can integrate Google Generative AI (GenAI), which provides intelligent and context-aware responses. This allows the system to handle complex queries and deliver more personalized interactions.

Furthermore, blockchain technology can be incorporated to ensure secure and transparent storage of transaction data. Each transaction is recorded as a block, preventing unauthorized modifications and ensuring data integrity. This enhances trust, reliability, and accountability within the system.

CHAPTER-7

IMPLEMENTATION

7. IMPLEMENTATION

The proposed system is a voice-enabled food ordering solution that utilizes advanced AI technologies such as Whisper and DistilBERT for accurate speech recognition and intent detection. It enables users to place orders through natural voice interactions, improving efficiency, reducing manual effort, and minimizing errors in communication. The implementation integrates speech processing, natural language understanding, and backend automation to ensure smooth system performance. A robust architecture supports order management, billing, and inventory control, along with secure transaction handling for data reliability. Additionally, the system includes analytics and reporting features to provide valuable business insights. Overall, the implementation delivers a scalable, reliable, and intelligent solution suitable for modern, technology-driven restaurant environments.

7.1 Environment and Core Dependency Initialisation

The implementation begins with setting up all necessary libraries and resources to ensure smooth operation. High-performance packages such as faster-whisper for STT and the transformers library for DistilBERT are installed, along with FFmpeg for reliable audio processing. The Gemini API Key is initialised to enable the gemini-2.5-flash model for advanced multilingual NLP. Persistent storage directories like order_data and daily_reports are created, with all_orders.json serving as the main ledger. The Whisper (faster-whisper) model is loaded with automatic device selection and optimized precision for fast, accurate, multilingual audio transcription.

7.2 Robust Data Modelling and Transactional Logic

The system defines MENU_DATA for items, pricing, stock, and popularity, while RESTAURANT_INFO ensures GST and legal invoice compliance. ITEM_MAPPINGS supports natural language parsing by mapping colloquial or phonetic references to standard menu names, and NATIVE_COMMAND_MAP enables multilingual command translation. The OrderManager class handles the full order lifecycle: add_to_order() updates quantities and suggests alternatives, remove_from_order() modifies the order, and finalize_order() calculates subtotal, GST, and total payable while preparing structured invoice data.

7.3 Hyper-Efficient Voice AI and Conversational Pipeline

Voice input is transcribed using `speech_to_text`, then classified with `DistilBERT` for intents like `ADD_ORDER` or `FINALIZE_ORDER`, allowing simple tasks to bypass Gemini and reduce latency. Complex commands are processed through Gemini NLP with hint-optimized prompts, producing structured instructions such as “add: Pepperoni Pizza:2.” `text_to_speech` generates natural audio responses in the customer’s language, and `process_gemini_commands` executes system actions for both single and multi- step interactions.

7.4 Auditable Financial Reporting and Blockchain Simulation

Invoices are generated in PDF and HTML formats with detailed itemization and GST. Transactions are stored in `all_orders.json` with timestamps, and a blockchain simulation maintains immutable logs via `update_stock_on_chain()` and `finalize_order_on_chain()` for inventory and finalized orders. The Analytics Engine (Pandas and Plotly) produces visual insights on revenue trends, top-selling items, category-wise revenue, and KPIs.

7.5 Gradio Interface Design and Event Handling

The Gradio interface integrates all backend modules, offering separate login flows for customers and owners. Customers interact via a two-column layout with Menu Panel and Voice/Chat Order Panel, and a real-time Order Summary. Owners access blockchain-linked stock controls, analytics charts, and KPIs with live updates. Event handling ensures voice inputs trigger the `voice_chat_interface`, order finalization initiates invoice generation, and owner actions update inventory and analytics seamlessly.

In conclusion, the implementation combines voice recognition, natural language processing, and backend management to build a smooth and efficient food ordering system. Fast speech processing, accurate intent detection, and secure order handling help the system work reliably in restaurant environments. Overall, the system is simple to use, efficient, and suitable for modern voice-based food ordering.

CHAPTER-8

TESTING

8 TESTING

Testing is the process of evaluating a software system to ensure that it works correctly and meets the required specifications. It helps identify errors and verifies that all components of the system function properly. In this project, testing ensures accurate speech recognition, order processing, and invoice generation.

Testing a complex application like the Talk2Invoice Restaurant system requires covering all its components: the Speech/Text-to-Command Logic (Gemini), the Order Management business logic, the Owner Dashboard/Data Management, and the Gradio UI Flow. Here is a comprehensive set of test cases for validation (Valid) and error/edge (Invalid) scenarios, organized by system component. The provided file is a Python script that defines a comprehensive Voice-Controlled Restaurant Ordering System using Gradio, Gemini, and several other libraries for Speech-to-Text (STT), Text-to-Speech (TTS), and PDF generation. Testing this system requires a multi-faceted approach, covering its various components: the Speech/Text-to-Command Logic, the Order Management System, the Gradio UI Flow, and the Owner Dashboard/Analytics.

8.1 Unit Testing Strategy

Since the core logic is contained in distinct functions and the OrderManager class, these should be tested independently of the Gradio UI and the external Gemini/Whisper models.

Component / Area	Test Description	Expected Result
Core Functions	Test menu search, item mapping, and FAQ responses.	Correct item found or meaningful message shown.
Order Management	Add, remove, and finalize orders including unavailable items.	Orders updated correctly; alternative suggestions shown.
Invoice Generation	Verify that PDF or HTML invoices are generated properly.	Invoice file create

Table 8.1: Unit Testing Strategy

8.2 Integration Testing Strategy

This involves testing the interaction between the Gradio interface, the Gemini model, and the core logic. Since Gemini is the command interpreter, these tests are crucial.

A. Gemini Chat Command Parsing (`gemini_chat process_gemini_commands`)

The goal is to test if Gemini correctly translates natural language into the required commands and if `process_gemini_commands` executes them properly.

Component / Area	Test Description	Expected Result
Gemini Command Processing	Check if Gemini correctly converts user speech into commands (e.g., “I want a pizza”).	System adds items to order correctly.
Speech-to-Text (STT)	Speak clearly to test speech recognition.	Voice converted to correct text.
Text-to-Speech (TTS)	Verify system’s voice output in reply to commands.	Clear and accurate voice response.
Language Handling	Switch between English, Hindi, or other languages.	System responds in selected language accurately.

Table-8.2: Integration Testing Strategy

8.3 UI Flow and Scenario Testing

These tests verify the end-to-end user experience and state management across different Gradio groups

Component / Area	Test Description	Expected Result
Customer Order Flow	Test login, placing order, reviewing cart, making payment, and downloading invoice.	Smooth navigation; correct totals and details displayed.
Owner Dashboard	Admin login, stock check, and viewing business analytics.	Dashboard displays updated and accurate data.
Error Handling	Order unavailable or invalid items.	“Item not found” message or alternate suggestion displayed.

Table-8.3: 3UI Flow and Scenario Testing

8.4 Scenario Testing

scenario testing means testing the system using real-world situations to make sure the application behaves correctly when users perform normal activities. It helps improve software quality by validating the overall functionality and usability of the system.

Scenario	Test Description	Expected Result
Customer Order	Order “1 pizza and 1 coke” and finalize it.	Order processed correctly; PDF invoice generated.
Out-of-Stock Item	Try ordering unavailable item (e.g., Vanilla Ice Cream).	System suggests available alternatives.
Language Change	Switch to Hindi and give a voice command.	Command recognized and response in Hindi.
Admin Functionality	Admin views top-selling items and generates daily report.	Report created with accurate sales and revenue data.

Table-8.4: Scenario Testing

In conclusion, the testing process confirmed that all components of the Talk2Invoice system work correctly, including voice recognition, order management, invoice generation, and dashboard functions. The results show that the system handles both normal and error cases properly, ensuring reliable performance and a smooth user experience.

CHAPTER-9

RESULTS AND ANALYSIS

9 RESULT AND ANALYSIS

9.1 Application Working Snapshots

Login Page

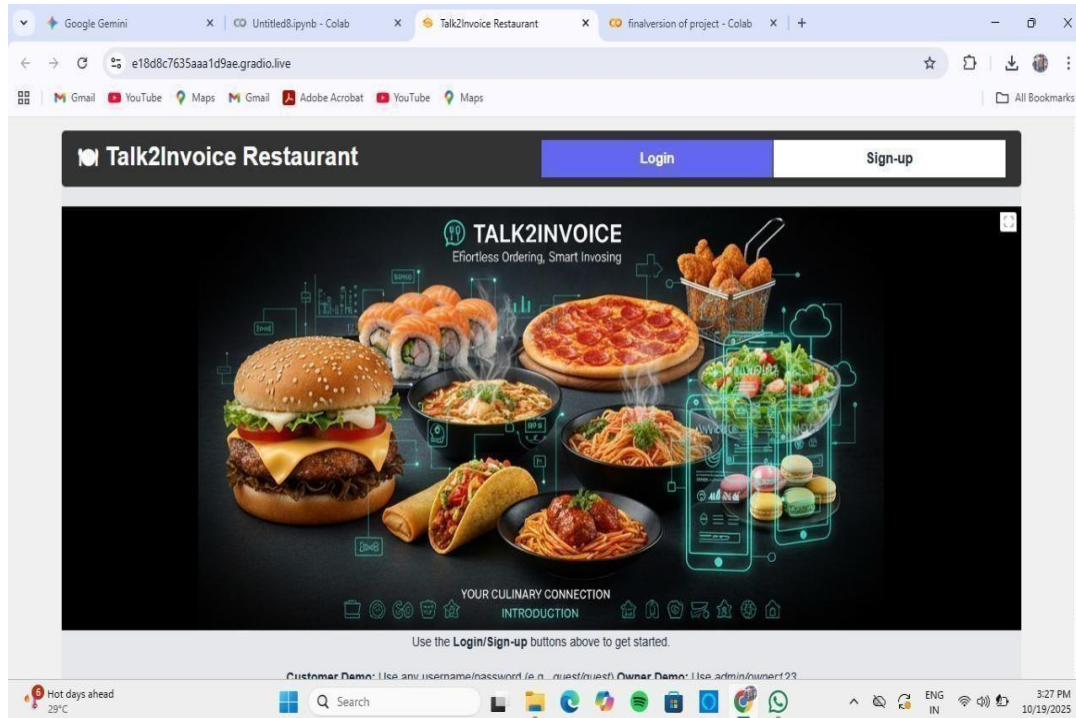


Fig. 9.1: Login Page

Menu & Order Placing

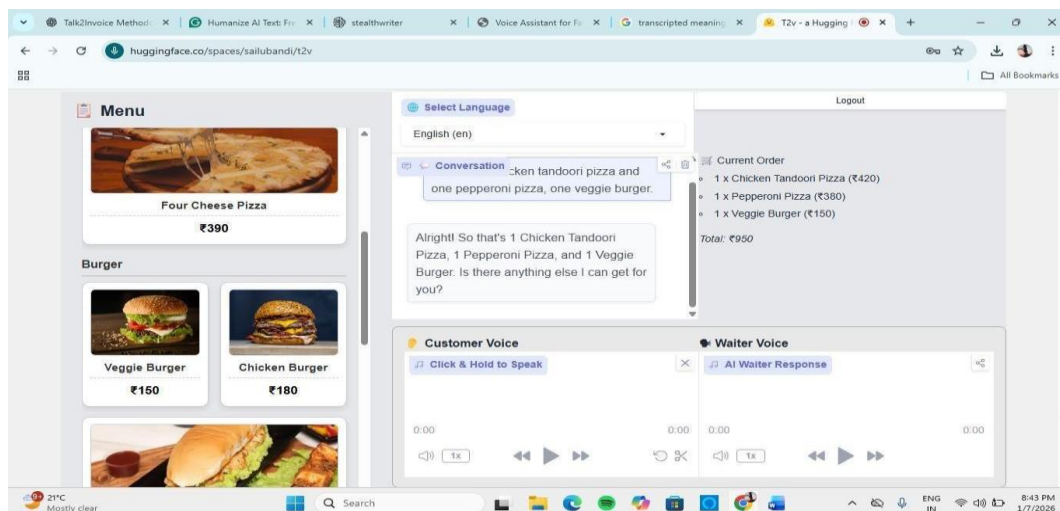


Fig. 9.2: Menu & Order Placing Page

Confirming Order

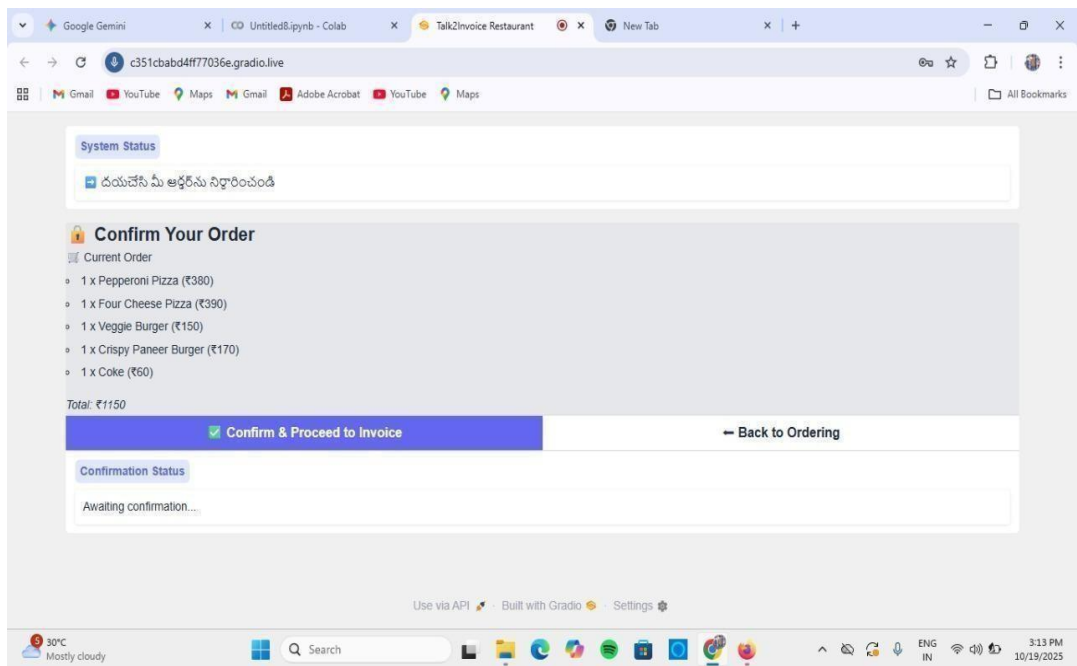


Fig. 9.3: Confirming Order

Invoice Preview

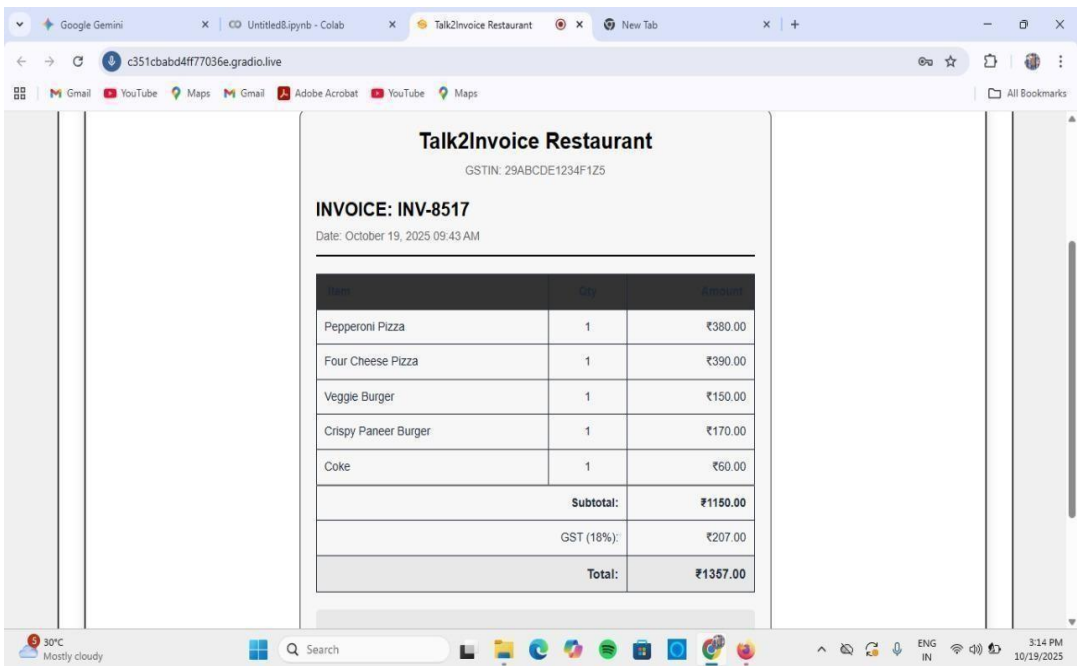


Fig. 9.4: Invoice Preview

Admin Dashboard

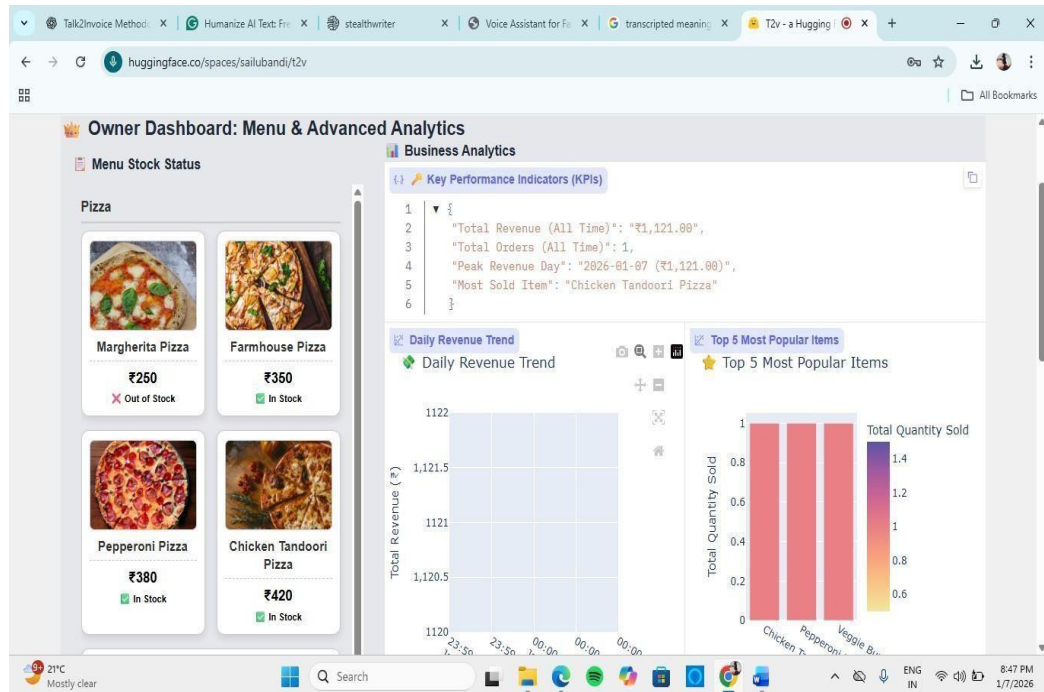


Fig. 9.5: Admin Dashboard

Revenue by Category

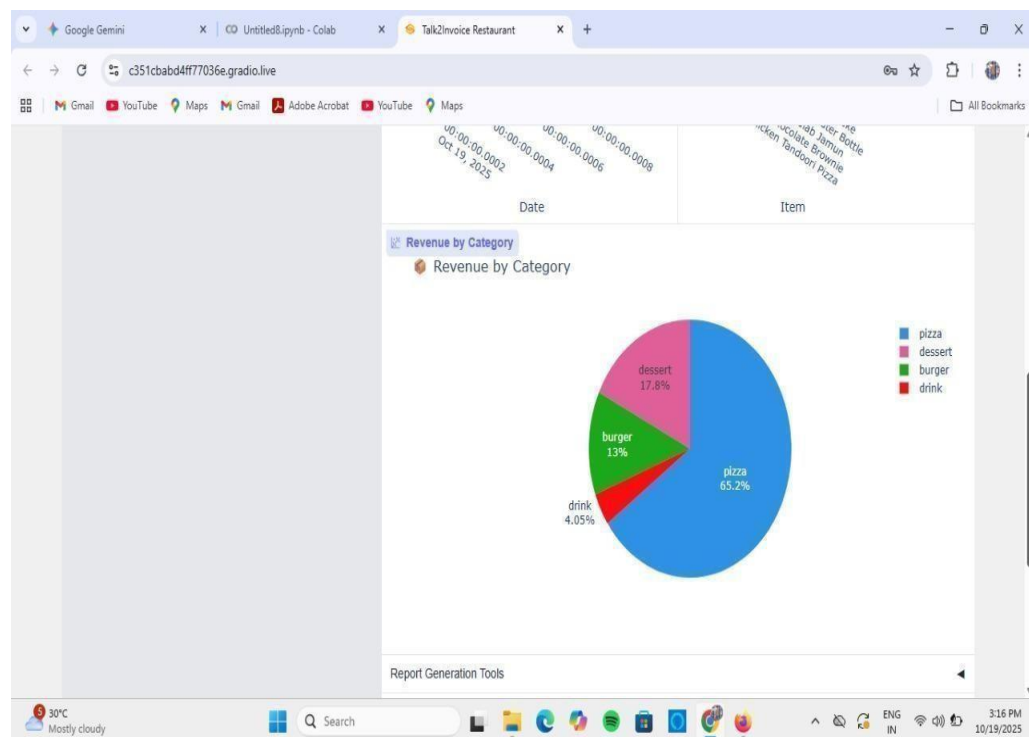


Fig. 9.6: Revenue by Category

9.2 Model Accuracy of the Intent Recognition Model

We tested intent recognition model in terms of its capacity to find the correct interpretation of spoken food orders. Each input produces a single prediction of the model. The validation data were used to determine the accuracy and compare the predicted outputs against the true labels.

Throughout the training, the DistilBERT based classifier displayed consistent improvement and an accuracy of approximately 96% on the validation set. The trend of accuracy during training is that there is steady improvement with slight fluctuation in the final stages of training, as a pointer of good learning and good generalisation. This step of user confirmation will also contribute to the increased system reliability by making sure that no wrong orders can be confirmed.

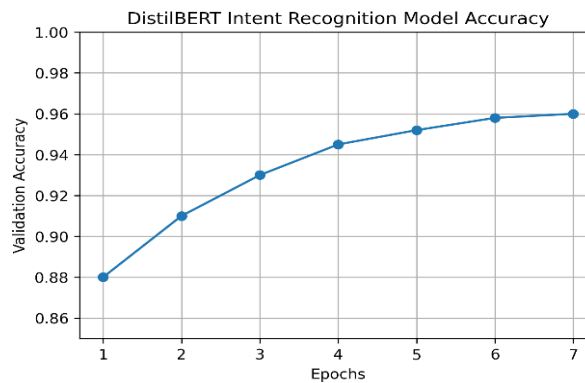


Fig. 9.7: *The validation accuracy of the DistilBERT intent order recognition model vs. training epoch.*

9.3 Practical order Scenario validation

The practical applicability of the proposed system was evaluated using real-world restaurant ordering scenarios involving both single-item and multi-item verbal requests. The evaluation focused on assessing the system's ability to accurately interpret natural language inputs and convert them into structured order data.

Order ID	Verbal Request	System Interpretation	Status
O01	One Pizza and two cold drinks	Pizza ×1, Cold Drink ×2	Successful
O02	Two fries and a coke	Fries x2, Coke x1	Successful
O03	Chicken 3 burgers	Burger x3	Successful

Table 9.1: *Order validation*

Furthermore, the backend validation mechanisms effectively prevented the processing of unavailable or invalid items and ensured that generated orders aligned with operational constraints such as inventory availability and kitchen workload. The consistent success across all test cases confirms the system's robustness, reliability, and suitability for deployment in real-world restaurant environments.

9.4 Administrative Visibility and Transaction Logging.

Once an order is confirmed, all order details and transactions are stored in the backend database, which is secured. Also, transaction information is stored in a blockchain-based registry, which is secure, traceable, and transparent.

A dashboard in the administration provides real-time access to orders being worked on and orders completed, inventory levels, kitchen workload, and transactions. This centralised monitoring facility aids in good supervision, auditing and operational control both in normal and peak hours.

In conclusion, the results demonstrate that the Talk2Invoice system performs accurately in intent recognition, order validation, and real-time invoice generation under practical conditions. The analysis confirms that the system is reliable, efficient, and suitable for real-world restaurant environments, providing smooth operation even during multiple and complex orders.

CHAPTER-10

DEPLOYEMENT

10 DEPLOYMENT

The following describes the deployment of the Talk2Invoice system in a cloud environment to ensure scalability, accessibility, and reliable performance. The application is hosted on Hugging Face Spaces, enabling efficient execution of AI models with support for both CPU and GPU resources. It also covers model management, dependency configuration, and secure handling of system components. Additionally, automated updates and continuous integration practices are used to maintain system stability and performance. Overall, the deployment ensures smooth, real-time operation and easy access for users.

10.1 Cloud Infrastructure and Hosting

The Talk2Invoice application is deployed on Hugging Face Spaces, a cloud-based platform designed for hosting machine learning and AI-powered applications. It provides a simple and efficient environment for deployment without requiring complex infrastructure setup, making it ideal for rapid development, testing, and real-world usage. The platform supports both CPU and GPU environments, offering flexibility based on computational requirements.

In this project, GPU support is utilised to efficiently run the Faster Whisper Speech-to-Text (STT) model, as speech recognition involves deep learning tasks that require high processing power. GPU acceleration significantly improves processing speed and reduces latency, enabling real-time voice interaction and a seamless user experience. The application runs within the Hugging Face runtime environment, where all required components such as Python dependencies, FFmpeg, and pre-trained models are properly configured.

Additionally, the platform provides automatic scaling and efficient resource management, ensuring consistent performance under varying workloads. It also simplifies deployment by handling backend infrastructure, allowing developers to focus on application logic. The system is accessible through a web interface, enabling easy testing, demonstration, and real-world interaction.

10.2 Model and Dependency Management

Efficient model and dependency management ensure the smooth execution of the Talk2Invoice system. The Whisper model is loaded at runtime to process voice inputs with minimal delay, improving response time and interaction quality.

All dependencies are defined in configuration files such as requirements.txt to maintain consistency and avoid compatibility issues. The project uses GitHub for version control and integrates with Hugging Face Spaces for automatic redeployment on updates. Sensitive data, including API keys, is securely managed using environment variables to enhance security and flexibility.

10.3 CI/CD and Stability

The Talk2Invoice system follows a simplified Continuous Integration and Continuous Deployment (CI/CD) approach using GitHub and Hugging Face Spaces to ensure continuous improvement and stability. Whenever code changes are committed and pushed to the GitHub repository, the application is automatically rebuilt and redeployed on Hugging Face Spaces, ensuring that the latest version is always available without manual intervention.

Before deployment, basic testing and validation are performed to ensure that new updates do not introduce errors or disrupt existing functionality, thereby maintaining system reliability. Hugging Face Spaces provides built-in logging and monitoring features that help track application performance, detect runtime errors, and analyse metrics such as response time and resource usage. The system is optimised to handle multiple user requests efficiently by reducing latency and improving model performance, and regular updates are carried out to enhance features, improve accuracy, and ensure secure and stable operation in real-world scenarios.

In conclusion, the deployment on Hugging Face Spaces with proper model management and CI/CD integration ensures that the Talk2Invoice system runs smoothly in a cloud environment. This setup provides stable performance, faster processing, and easy updates, making the system reliable for real-time restaurant usage.

CHAPTER-11

FUTURE SCOPE

11 FUTURE SCOPE

The **Talk2Invoice** system has a wide range of future improvements and advanced features that can make it even more intelligent, efficient, and user- friendly.

11.1 Offline Mode Support:

To improve reliability, the system can be enhanced with an offline mode that allows it to function even when there is no internet connection. In this mode, customer orders and interactions are temporarily stored in a local database or device storage. Once the internet connection becomes available again, the system automatically synchronises the stored data with the central server without requiring manual intervention. This feature ensures that restaurant operations continue smoothly even in locations with unstable networks, preventing order loss and maintaining service continuity.

11.2 Emotion Detection in Voice:

Future versions of the system can incorporate emotion detection technology using advanced speech analysis and artificial intelligence. By analysing elements such as tone, pitch, speech speed, and voice intensity, the system can identify the emotional state of the customer (e.g., happy, frustrated, confused, or neutral). Based on this analysis, the system can adjust its responses to create a more natural and empathetic interaction. For example, it may respond calmly and offer assistance when detecting frustration, or use a more cheerful tone when the customer sounds happy. This feature would enhance user experience and make voice interactions feel more human-like.

11.3 Integration with Delivery Partners:

The system can be integrated with popular online food delivery platforms such as Swiggy and Zomato to automate the ordering and delivery process. When a customer places an order through the voice system, the order details can be automatically transmitted to the delivery platform. The system can then handle order dispatch, delivery partner assignment, and real-time tracking. Customers may also receive voice updates regarding order confirmation, preparation status, and estimated delivery time. This integration would streamline restaurant operations and customers with ordering .

CHAPTER-12

CONCLUSION

12 CONCLUSION

The Talk2Invoice system is an advanced AI-based voice ordering solution that transforms traditional restaurant operations into a smart, contactless, and fully automated process. It integrates Artificial Intelligence (AI), Natural Language Processing (NLP), Speech Recognition, and Blockchain Technology to deliver a fast, secure, and intelligent customer experience. Whisper and Fast Whisper provide accurate speech-to-text conversion even in noisy environments, while DistilBERT accurately interprets user intent and menu selections. The backend automates menu management, inventory control, and kitchen operations for real-time tracking, error reduction, and optimized efficiency. Digital invoices are generated automatically using ReportLab, and blockchain integration ensures secure, tamper-proof, and transparent transaction records, building trust between customers and restaurant owners. The admin dashboard offers insights on daily sales, top-selling items, and performance analytics to support data-driven decisions. Customers benefit from multi-language voice support, fast ordering, and accurate billing, improving satisfaction and reducing wait times. The system reduces human involvement, minimizes staff workload, and enhances overall service speed. Additionally, Talk2Invoice contributes to eco-friendly practices by replacing paper receipts with digital invoices and automating backend processes. Demonstrating how AI, automation, and blockchain can create a reliable restaurant ecosystem, the system can be further expanded with features such as offline mode, emotion-based responses, AI-powered feedback, voice-controlled kitchen monitoring, and robotic order delivery. Overall, Talk2Invoice represents a significant step toward next-generation, AI-powered, voice-interactive restaurant systems, setting a benchmark for innovation in the food industry.

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Talk2Invoice: AI Powered Voice Assistant for Food Ordering and Invoicing

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Abstract

Artificial intelligence is being adopted more in fast-food service environments to make ordering quicker and easier, especially in busy places like metro stations, airports, and self-service kiosks. However, many current solutions don't meet expectations due to limited voice recognition, lack of real-time inventory updates, and inadequate backend systems. Talk2Invoice is a voice assistant powered by AI that aims to solve these issues by allowing contactless food ordering through natural speech and automating backend tasks. Multilingual voice input is turned into text using OpenAI's Whisper model, and customer intent along with food item details is identified using a fine-tuned DistilBERT model. QR code-based menu access makes interactions simpler, while a blockchain-enabled backend supports real-time stock management, queue optimization, and secure digital transactions. Once the order is confirmed successfully, digital invoices are automatically generated and sent to users, and analytical reports provide insights into sales trends and service performance. Experimental results show improved order accuracy in noisy settings, and better transparency in transaction handling. By combining speech processing, natural language understanding, and secure backend technologies, Talk2Invoice presents a practical, scalable, and user-friendly method for AI-driven food ordering in modern service environments.

Keywords: *Voice Assistant, Speech-to-Text, Whisper (Faster Whisper), DistilBERT, Natural Language Processing, Real-Time Inventory Management, Digital Invoicing; Blockchain.*

1. Introduction

Artificial intelligence is reshaping service industries by enabling smart automation and natural human-computer interaction. In fast-food and quick-service restaurant settings, especially in busy places like airports, metro stations, food courts, and self-service kiosks, efficient order placement and transaction handling are crucial for maintaining service quality. Traditional food ordering methods, such as manual counters and touchscreen kiosks, often lead to long lines, order mistakes, and limited scalability during peak hours. These problems have increased the demand for AI-driven solutions that can improve operations and enhance the customer experience.

Voice-based interfaces provide a convenient and contactless option for food ordering. However, current voice assistants often struggle in real-world fast-food environments due to background noise, multilingual interactions, and limited connections to backend systems. These issues frequently cause incorrect intent interpretation, delayed responses, and inconsistent order fulfillment. To tackle these challenges, this study presents Talk2Invoice, an AI-powered voice assistant designed to automate food ordering and invoicing through natural speech interaction, focusing on reliability, scalability, and transparency.

1.1 The Need for Voice-Based Food Ordering Automation

Fast-food environments need quick and precise order processing to handle high customer volumes effectively. Manual and touchscreen systems often cause delays and errors, especially during busy times. Additionally, the growing demand for contactless interactions has made it necessary to have ordering methods that minimize physical contact and are easy to use. Voice-based food ordering offers a natural solution, but its effectiveness relies on strong speech recognition and seamless backend integration. There is a clear need for intelligent voice-driven platforms that combine ordering, inventory checks, and billing into a single automated workflow.

1.2 Challenges in Existing Voice-Based Ordering Systems

Current food ordering systems, including voice-enabled solutions, face several practical challenges:

1. Limited Multilingual Support: Insufficient handling of various languages limits accessibility.
2. Poor Backend Integration: A lack of real-time connections with inventory and billing systems causes order inconsistencies.
3. Latency and Reliability Issues: Delayed responses and intent errors decrease user confidence.

These challenges hinder widespread adoption and show the need for a stronger AI-based approach.

System Architecture

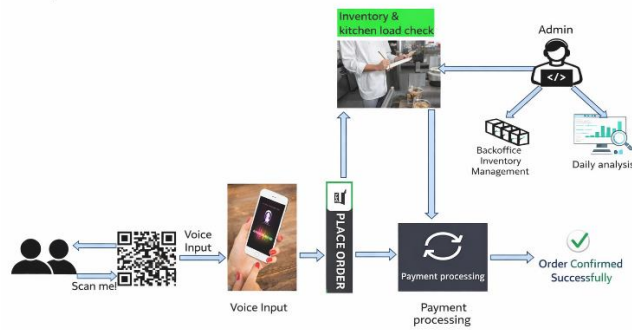


Fig-1: System Architecture of the Talk2Invoice Platform

1.3 Role of Whisper and DistilBERT in Voice Understanding

Talk2Invoice addresses challenges in speech and language understanding by using advanced deep learning models. OpenAI's Whisper model assists with multilingual speech-to-text conversion and works well even in noisy settings. A fine-tuned DistilBERT model helps with accurate intent recognition and food item extraction from user commands. Together, these models enable reliable voice understanding, while backend automation supports real-time inventory checking, kitchen load balancing, and digital invoice creation.

1.4 Objectives of the Study

The objectives of this study are:

1. To develop a strong voice-based food ordering solution for noisy and multilingual environments
2. To integrate real-time inventory checks and automated digital invoicing
3. To ensure secure and transparent transactions using blockchain technology
4. To assess improvements in order accuracy and response time

1.5 Significance of the Study

This research offers a practical AI-driven solution to key challenges in fast-food operations. By automating ordering and invoicing through voice interaction, Talk2Invoice decreases waiting times, reduces human errors, and improves service consistency. The combination of speech processing, natural language understanding, and secure backend technologies enhances transparency and operational reliability. For service providers, the platform reduces staff workload and enables data-driven decision-making, while customers enjoy a convenient, contactless, and multilingual ordering experience. Overall, this study showcases the potential of AI-powered voice

assistants to support scalable and user-friendly food ordering systems in modern service settings.

2. LITERATURE SURVEY

According to the recent studies, there is an increasing trend of applying artificial intelligence to voice-based ordering and the automation of the food service. Srinivasan et al. [1] show that the accuracy of intent recognition in fast-food voice assistants can be dramatically increased by the use of transformer-based NLP models, in particular, DistilBERT. Their study validated the idea that deep learning techniques can be useful in understanding customer orders and provided the basis of the continued evolution of AI-based food ordering systems.

Kim et al. [2] suggested a voice assistance framework that centers on deep learning and emphasizes on speech preprocessing, contextual interpretation and noise suppression. In their research, they indicated that consistent speech-to-text systems are essential to make them work in a real restaurant environment. On the same note, Zhang et al. [3] proposed the implementation of an intelligent food ordering system that includes speech recognition with semantic analysis, which additionally demonstrates the potential and the advantages of AI-based ordering systems.

The potential of voice interfaces to automate restaurants was demonstrated by Shirode et al. [4], who developed a speech-based virtual restaurant clerk based on the Web Speech API and NLP, and the authors of the study. Later, Kumar et al. [5] retrospected the development and use of voice assistants, and usability, system integration, and reliability were cited as such a problem in the real world. Ashfaq et al. [6] also examined the style of chatbot interaction in online food delivery and emphasized that the conversational design and the quality of interactions are the major factors of user engagement and satisfaction.

Patil et al. [7] introduced a voice-assisted online food ordering web-application, which demonstrated excellent practicability and convenience but not secure automated support on the backend and lacked mass evaluation. Rathod et al. [8] also experimented with speech or voice-based restaurant assistants, and found that voice-based ordering and billing worked successfully. This, however, came with constraints on scalability, adaptability and robustness in noisy real world situations due to their dependence on browser-based technologies and predefined dialogue structures.

Much more recently, Huang et al. [9] have suggested a hybrid on-device food ordering system which uses lightweight language models and rule-based elements but focuses on the concern of resource-bound environments and efficiency/scalability. Based on these works, the proposed Talk2Invoice system seeks to address the weaknesses of robustness, end-to-end automation, transformer-based intent

understanding, secure blockchain-based invoicing, and analytics-based management support.

3. Methodology

This section presents the process of designing and implementing the solution to the voice-based system of ordering and invoicing food, Talk2Invoice. The method consists of three key elements, which include voice interaction and intent processing, order processing and validation, and backend and administrative management. This systematic methodology shows the functioning of the system since interaction with the customers up to the control side and is a sure way of reliability and efficiency in restaurants.

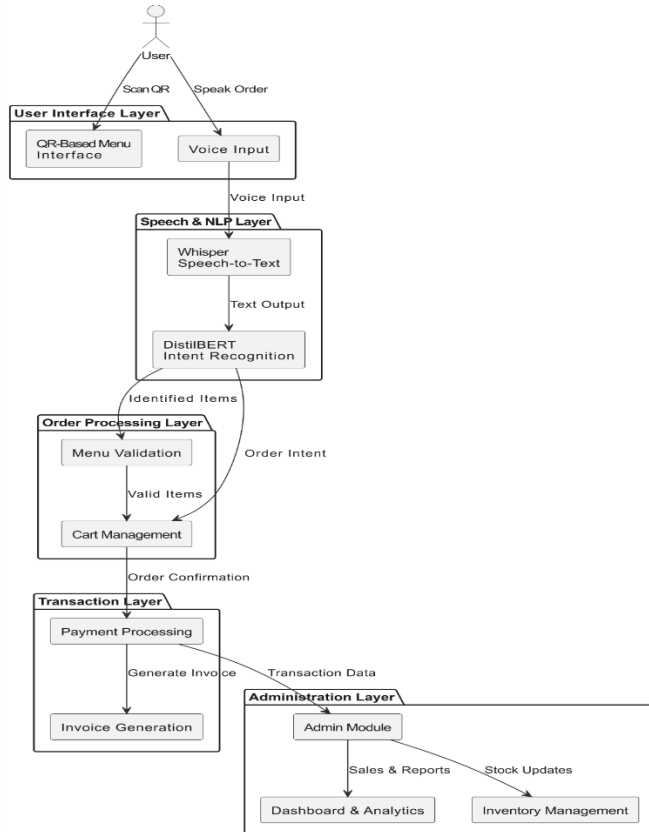


Fig-2: Methodology flow chart

3.1 voice interaction and intent processing process.

This begins with a customer scanning a QR code in the restaurant which directs to a digital menu interface. The interface is browser-based and is easy and it allows the customers to use the system without the need to install an app. There is also an interactive element of voice in the menu.

The customer orders a product after talking to a device that has a microphone. Background noise or other speech patterns may be picked up by the recorded voice which is common in restaurants. This sound is sent to a Speech-to-Text (STT) engine which translates the request to text using the Whisper or Faster-Whisper engine.

Through the Intent Recognition module, the text is then run through a fine-tuned DistilBERT model. This model examines the context in order to determine what the user wants and extract appropriate information such as the name and quantities of menu items. The information that was interpreted is added to the cart temporarily and a voice confirmation message is issued by the system to enable the customer check the detected order before proceeding.

3.2 Processing and validation of orders.

Once the user affirms with the order observed, the system initiates the order processing stage. The backend performs feasibility checks prior to the acceptance of the order to make sure that nothing will go wrong.

The system determines the inventory first to identify whether the required items in stock. Meanwhile, it evaluates the workload of the kitchen so that the delays are not long, in particular, in peak times. Only the orders which comply with both the inventory and kitchen workload requirements are further processed. After the validation, the order is completed and assigned with a unique identifier, which aids in the tracking and reference during the life of the order.

3.3 Administrative and Backend Management.

Once the order is confirmed, the backend system safely stores the details of the order in database to track and keep the records of the order. Also, order and transaction metadata are stored in a blockchain-based ledger which ensures secure storage, immutability and transparency of records. This configuration facilitates in auditing, traceability and dependable recording of transactions. Once the order has been processed, a digital invoice is generated, which contains the details of the items and their quantities and prices and references of transactions. This is e-mailed to the customer.

There is an administrative monitoring interface where the restaurant personnel can monitor the manner in which the systems operate. With this interface, administrators are able to monitor the orders placed and orders fulfilled, control stored inventory, monitor kitchen workload and finally check on the transaction record. This centralized management facilitates proper control and proper management of the restaurant.

3.4 Model Training and Evaluation.

Talk2Invoice system has been trained and tested through a comprehensive method to make sure that the intent recognition is correct and the performance is reliable in real-time.

Model Training

The intent recognition component was realized with the help of a fine-tuned DistilBERT model that was trained on a selection of food item descriptions and utterances related to orders. The DistilBERT tokenizer was used to tokenize the text and AdamW

was used to train with a batch size of 32 and a range of learning rates between 2×10^{-5} to 5×10^{-5} . Monitoring validation accuracy and loss during training was done and to prevent overfitting early stopping was used.

Inference and Post Processing

In inference the Whisper speech recognition model was used to first translate spoken user input into text. The output of that transcription was fed into the trained DistilBERT classifier to determine the intent of the user and the types of foods they wanted, which were converted to backend operations (order verification and invoice creation).

Model Evaluation

Accuracy and Macro-averaged F1-Score were used in evaluating model performance over a held-out test set. The intent recognition module had a maximum accuracy of 96.30%.

4. Experimental Results and Analysis

This part includes the discussion and analysis of the Talk2Invoice system experimentally. The analysis is based on the overall process of operation with emphasis to voice-based order placement, accuracy in intent recognition, backend validation, transaction processing, and administrative transparency. The findings indicate that the system is efficient in food ordering and invoicing in restaurants in real time.

4.1 System Workflow and Functional Results

The Talk2Invoice system has a definite workflow of user interaction to the invoice generation. To test the accuracy and reliability of the system, several order scenarios were used to test the system. Customers use a QR code to access the platform and see the menu in its digital format and make orders via natural spoken language. The voice input is transcribed into text through the Whisper-based Speech-to-Text engine which works effectively in a restaurant setting with moderate background noises.

The transcribed text is passed through a model that is based on DistilBERT to give a final prediction where the menu items and quantities are identified. The identified order is provisionally put in the cart and verified through the voice reply. Then, inventory availability and load in the kitchen are verified by backend services. Passing orders are finalized, assigned a unique order identifier and then digital invoices are generated.

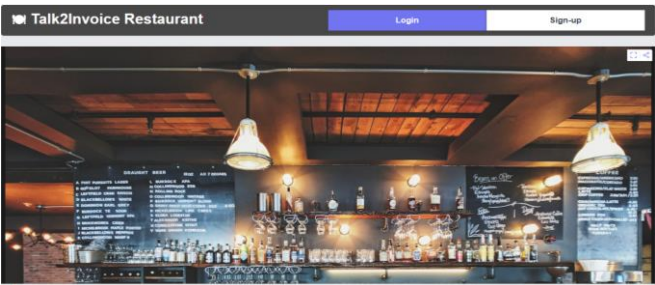


Fig-3: The Talk2Invoice Interface

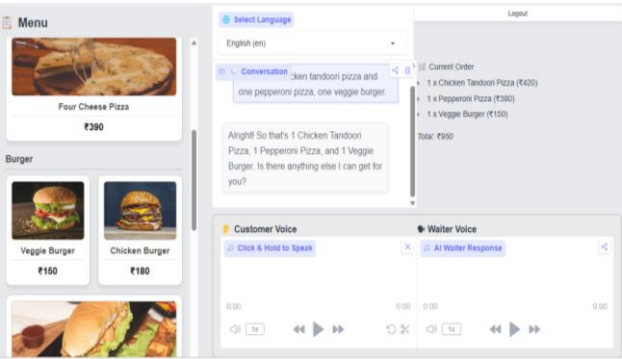


Fig-4: The Talk2Invoice Ordering Page

4.2 Model Accuracy of the Intent Recognition Model

We tested intent recognition model in terms of its capacity to find the correct interpretation of spoken food orders. Each input produces a single prediction of the model. The validation data were used to determine the accuracy and compare the predicted outputs against the true labels.

Throughout the training, the DistilBERT based classifier displayed consistent improvement and an accuracy of approximately 96% on the validation set. The trend of accuracy during training is that there is steady improvement with slight fluctuation in the final stages of training as a pointer of good learning and good generalization. This step of user confirmation will also contribute to the increased system reliability by making sure that no wrong orders can be confirmed.

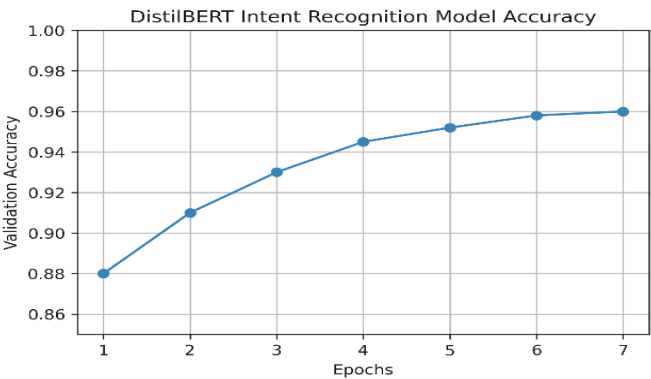


Fig-5: The validation accuracy of the DistilBERT intent order recognition model vs. training epoch.

4.3 Practical order Scenario validation

We evaluated the practical applicability of the system by testing the real-life application of the system to real restaurant order cases involving single-item and multi-item order requests.

Table 1: Order validation

Order ID	Verbal Request	System Interpretation	Status of the order
O01	One Pizza and two cold drinks	Pizza ×1, Cold Drink ×2	Successful
O02	Two fries and a coke	Fries x2, Coke x1	Successful

O03	Chicken burgers	3	Burger x3	Successful
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The backend validation modules were successful in disallowing unavailable items to be processed and designed a workable kitchen workload. Such findings affirm the validity and accuracy of the order processing system in the real situation.

4.4 Administrative Visibility and Transaction logging.

Once an order is confirmed all the details of order and transactions are stored in the backend database which is secured. Also, transaction information is stored in a blockchain-based registry, which is secure, traceable, and transparent.

A dashboard in the administration provides real-time access to orders being worked on and orders completed, inventory levels, kitchen workload, and transactions. This centralized monitoring facility aids in good supervision, auditing and operational control both in normal and peak business hours.

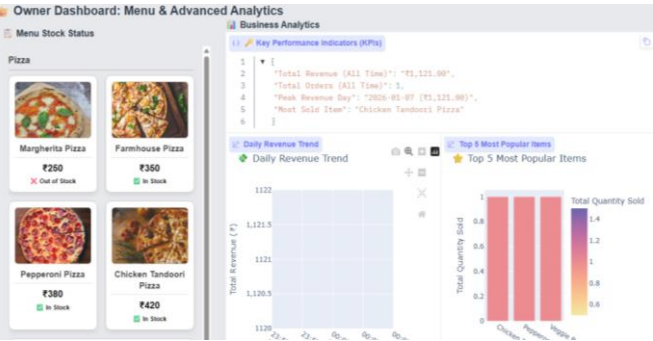


Fig-6: The Talk2Invoice Owner Dashboard

Talk2Invoice Restaurant		
GSTIN: 29ABCDE1234F1Z5		
INVOICE: INV-8517		
Date: October 19, 2025 09:43 AM		
Pepperoni Pizza	1	₹380.00
Four Cheese Pizza	1	₹390.00
Veggie Burger	1	₹150.00
Crispy Paneer Burger	1	₹170.00
Coke	1	₹60.00
Subtotal:		₹1150.00
GST (18%):		₹207.00
Total:		₹1357.00

Fig-7 The Talk2Invoice Generated Invoice

5. Discussion

This part will include the findings of the Talk2Invoice system and the success of the system in the actual restaurant environments. It presents some key findings, advantages of the voice approach. The findings indicate that Talk2Invoice is a stable, effective, and convenient system to use in automated food ordering and invoicing.

5.1 Major Findings and Implications.

The findings suggest that Talk2Invoice manages to automate the whole process of restaurant ordering by means of voice interaction. This system is a combination of speech-to-text conversion, intent recognition, backend validation, and invoice generation. The intent recognition model implemented using the DistilBERT model has a validation accuracy of approximately 95 to 96, which can be interpreted as high capability in comprehending the intent of the users correctly and recognize sequential items and quantities in natural language. The good generalization with the various ordering patterns and the smooth training process indicates stable learning.

Operationally, combining inventory verifications and kitchen load authentication are a huge reduction in the number of order rejection and preparation delays. These results suggest that voice ordering system using AI has the potential to increase the service quality and reduce the need to enter orders manually, particularly during the peak time in restaurants.

5.2 Benefits of the Proposed System.

Talk2Invoice system has a number of advantages compared to the traditional menu-based or manual ordering systems. It lowers significantly the time and effort required by customers to make orders since they can do it through their voice. This is an intuitive type of communication that is not complicated even by users with different backgrounds and technology. The intent recognition is based on AI and guarantees a high level of accuracy, mitigating such frequent errors as the wrong choice of item or a wrong amount of items.

Also, real-time backend verification improves the efficiency of operations since the lack of a certain product in the menu does not lead to the order, and the workload of the kitchen is not increased in advance. Additional usability is also facilitated by the administrative dashboard which gives continuous access to orders, inventory, and records of transactions. Preservation of secure transaction logging ensures that audits and performance reviews are possible due to the transparency and traceability. In general, the design of the system is modular and scalable, which allows it to be used in both small restaurants and large food service chains.

6. Future Directions

The system analysis and the experimental evaluation demonstrate that the Talk2Invoice may be used as a voice-based service to order food. It is an integrated Whisper speech recognition, DistilBERT intent comprehension, and automated backend processing. Although our current prototype is good in a controlled environment, we still require upgrade on scaling, reliability and preparedness in application in the real world. The scope of future research will be to increase the size of training

data, to streamline the efficiency of models and to come up with lightweight deployment strategies in large-scale food service settings.

6.1 You-Smart Personalization and Recommendation Engine.

The next generation of Talk2Invoice can be developed with intelligent personalization and recommendation as the new features to enhance customer interest and performance. The system will be able to offer customized meals, combo deals, and more specific suggestions based on historical order information, time trends, and trendy menu items. Such features may result in customer satisfaction, repeat use, and assist restaurant operators in demand forecasting, menu enhancement, and promotion.

6.2 Improving Conversation Intelligence and Personalization.

Subsequent versions of Talk2Invoice can go beyond the simple commands to more intelligent conversational intelligence. We will be able to implement context aware dialogue management to facilitate the use of multi-step and conversational ordering tasks. The system should be able to adapt to the user, which is possible by including sentiment and emotion analysis based on the vocal cues. Individual based recommendation engines with user preferences and eating patterns can even enhance system intelligence and involvement.

6.3 Security, Privacy and Ethical Implementation.

As the Talk2Invoice deals with sensitive user data, such as voice records and transaction records, the future work should be on data security, the protection of privacy, and the ethical utilization of AI. This entails the use of secure data encryption, access controls, and adherence to the laws of data protection. Such aspects of ethics as transparency of AI-related decisions, informed user consent, and minimization of bias across languages and accents will play a critical role in the implementation of the system responsibly and reliably.

7. Conclusion

This paper discussed the use of artificial intelligence in automated voice-based food ordering, and showed how AI systems can be used to improve the accuracy, efficiency and scalability of restaurant operations. The proposed Talk2Invoice system is a combination of the Whisper and Fast Whisper models of multilingual and noise-resistant speech recognition and a fine-tuned DistilBERT model of intent recognition. On experimental implementation, it is found that the architecture can be used to support real time, hands free interaction that is suitable in high traffic food service settings.

One of the main contributions made by this work is that it resulted in the creation of end-to-end intelligent driver-involving a voice input and backend automation, involving inventory validation, order management, and automatic invoice

creation. The blockchain-based backend is used to guarantee safe, transparent, and immutable records of transactions, whereas analytics dashboards allow continuously monitoring and making decisions based on data.

The applications are not limited to basic ordering, which can be used to contribute to smart food service infrastructures. The system will help lower manual workload, shorter service delivery times, and customer confidence due to frequent order processing and effective e-invoicing. However, despite the promise of this, more research is needed. Future research will also be on multilingual dataset growth, lightweight deploying of edge, adaptive and privacy conscious intelligence and mechanisms. The research forms a good basis in the development of the next generation AI-based restaurant automation.

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